

gcd $\rightarrow$ Greatest Common Divisor

$\downarrow$
HCF $\rightarrow$ Highest Common Factor

a, b

$\text{gcd}(a, b) \Rightarrow x$ [greatest no. / factor which divides
both a & b]

$\downarrow$

$x \rightarrow a \% x == 0 \quad \& \quad b \% x == 0.$

x is highest such no.

$\text{gcd}(15, 25) \Rightarrow 5$

$\downarrow \quad \downarrow$
1 1
3 5
5 25
15

$\text{gcd}(12, 30) \Rightarrow 6$

$\downarrow \quad \downarrow$
1 1
2 2
3 3
4 5
6 6
12 10
15
30

$\text{gcd}(10, -25) \Rightarrow 5$

$\downarrow \quad \downarrow$
1 1
2 5
5 25
10

$-25 \% 5 = 0$

$-25 \% -5 = 0$

$\text{gcd}(0, 8) \Rightarrow 8$

$\downarrow \quad \downarrow$
1 1
2 2
3 4
4 8
5 16
6 24
7

$\text{gcd}(0, 10) \Rightarrow 10$

$\text{gcd}(0, -10) \Rightarrow 10$

$\downarrow \quad \downarrow$
1 1
2 2
3 3
4 4
5 5
6 6
7 7
8 8
9 9
10 10
11 11
12 12
13 13
14 14
15 15
16 16
17 17
18 18
19 19
20 20
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98 98
99 99
100 100

$\text{gcd}(-16, 24) \Rightarrow 8$

$\downarrow \quad \downarrow$
1 1
2 2
3 3
4 4
5 5
6 6
7 7
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97 97
98 98
99 99
100 100

8

Properties of gcd :- 11 $\rightarrow$ abs value

1) $\text{gcd}(a, b) = \text{gcd}(|a|, |b|)$

2) $\text{gcd}(a, b) = \text{gcd}(b, a) \rightarrow$ commutative

3) $\text{gcd}(0, x) = |x|$

if $x = 0$, $\text{gcd}(0, 0) \Rightarrow \alpha$ undefined

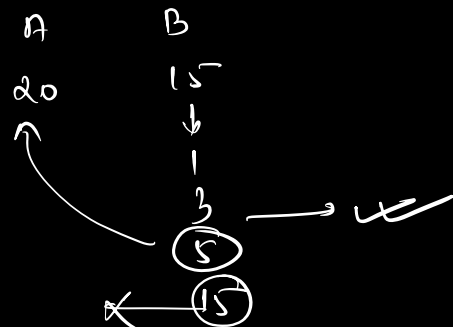
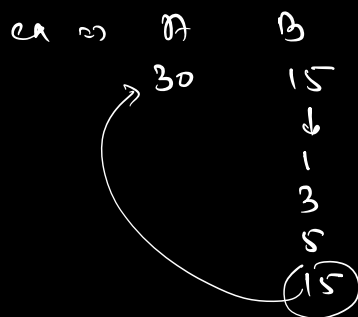
$\downarrow \quad \downarrow$
 $1 \quad 1$
 $2 \quad 2$
 $3 \quad 3$
 $\vdots \quad \vdots$

4) $\text{gcd}(a, b, c) = \text{gcd}(a, \text{gcd}(b, c))$
 $\text{gcd}(b, \text{gcd}(a, c))$
 $\text{gcd}(c, \text{gcd}(a, b))$

$\downarrow$
Associative

Q1. Given 2 nos. $A \neq B$, find $\text{gcd}(A \neq B) > 0$

$\Rightarrow$ soln 1



```
for ( i = min(A, B); i >= 1; i-- ) {
```

```
    if ( A % i == 0 & B % i == 0 ) {
```

```
        return i
```

```
    }
```

TC $\Rightarrow O(\min(a, b))$

SC $\Rightarrow O(1)$

for (i = min(A, B) $\xrightarrow{\text{iterate through all factors of } \min(a, b)}$

check for all

factors of $\min(A, B)$

```
}
```

TC $\Rightarrow O(\sqrt{\min(A, B)})$

SC $\Rightarrow O(\sqrt{\min(A, B)})$

A = 20 B = 15

15 x

14 x

13 x

12 x

11 x

10 x

9 x

8 x

7 x

6 x

5 x

$\Rightarrow$ Euclid Algo:

If a, b $a > b$

say, $\gcd(a, b) = x$

$a \% x = 0$

&

$b \% x = 0$

$\gcd(a-b, b) = x$

$(a-b) \% x = 0$

&

$b \% x = 0$

$(a \% x - b \% x + x) \% x$

$\downarrow$
0

$\downarrow$
0

$x \% x = 0$

if, $(a > b)$

$$\text{gcd}(a, b) = \text{gcd}(a-b, b)$$

Ex-2

$a > 0, b > 0$

$\text{gcd}(a, b) \{$

if $(a < b) \{$

swap (a, b)

$\}$

if $(b == 0)$ return a ;

return $\text{gcd}(a-b, b)$

$\}$

TC $\Rightarrow O(\max(A, B))$

$\text{gcd}(16, 1)$

↓

$\text{gcd}(15, 1)$

↓

$\text{gcd}(14, 1)$

↓

$\text{gcd}(13, 1)$

↓

0

$a=8, b=6$

$\text{gcd}(8, 6) \{$

return $\text{gcd}(2, 6)$

$\}$

$\text{gcd}(2, 6) \{$

swap

$a=6, b=2$

return $\text{gcd}(4, 2)$

$\}$

$\text{gcd}(4, 2) \{$

return $\text{gcd}(2, 2)$

$\}$

$\text{gcd}(2, 2) \{$

return $\text{gcd}(0, 2)$

$\}$

$\text{gcd}(0, 2) \{$

// swap

$a=2, b=0$

return $\text{gcd}(2, 0)$

$\}$

$$\begin{array}{l|l}
 17 - 5 = 12 & 17 \% 5 = 2 \\
 12 - 5 = 7 & \\
 7 - 5 = 2 &
 \end{array}$$

Ex

$$\begin{array}{ccccccc}
 \overset{a}{(23, \overset{b}{5})} & \rightarrow & \overset{a}{(18, \overset{b}{5})} & \rightarrow & \overset{a}{(13, \overset{b}{5})} & \rightarrow & \overset{a}{(8, \overset{b}{5})} \rightarrow \overset{a}{(3, \overset{b}{5})} \\
 & & & & & & \uparrow \\
 & & & & & & 23 \% 5
 \end{array}$$

$$(23, 5) \rightarrow (23 \% 5, 5)$$

$$\gcd(a, b) \rightarrow \gcd(a - b, b)$$

$$\gcd(a - 2b, b)$$

$$\gcd(a - 3b, b)$$

$$\begin{array}{c}
 \vdots \\
 \vdots \\
 \gcd(a - nb, b) \\
 \hline
 \downarrow
 \end{array}$$

$$a - nb \geq 0$$

$\downarrow$
 max multiple
 of $b \leq a$

$$\boxed{\gcd(a, b) \Rightarrow \gcd(b, a \% b)}$$

$$\boxed{a \% b} < b$$

```

gcd(a, b) {
    if (b == 0)
        return a
    return gcd(b, a % b)
}

```

$\text{gcd}(15, 6)$
 $\downarrow$ $15 \div 6$
 $\text{gcd}(6, 3)$
 $\downarrow$ $6 \div 3$
 $\text{gcd}(3, 0)$
 $0!p=3$

$\text{gcd}(6, 21) = 3$
 $\downarrow$ $6 \div 21$
 $\text{gcd}(21, 6)$
 $\downarrow$ $21 \div 6$
 $\text{gcd}(6, 3)$
 $\downarrow$ $6 \div 3$
 $\text{gcd}(3, 0)$
3

TC $\Rightarrow$

$$\text{gcd}(a, b) = \text{gcd}(a \% b, b)$$

$$b \leq a/2$$

$$a \% b \leq b$$

$$\rightarrow a \% b \leq a/2$$

$$\text{gcd}(a, b) \neq \text{gcd}(a \% b, b)$$

$$\text{gcd}(\leq a/2, b)$$

log

$$b > a/2$$

$$a \% b = a - nb$$

$$n \geq 1$$

$$a \% b = a - b \leq a/2$$

$$a \% b \leq a/2$$

$$n \left(\begin{matrix} > a/2 \\ \rightarrow \end{matrix} \right)$$

$$> a$$

$$n = 1$$

$$a - b > 0$$

$$n \geq 2$$

$$a - 2b > 0$$

$$TC \rightarrow O(\log(\min(a, b)))$$

Amazon

Q2. Given N arr elements, find the hcd of factorials of all given element

ex $\rightarrow$ arr $[4, 3, 8, 6]$

$$\begin{array}{cccc} & \downarrow & \downarrow & \downarrow & \downarrow \\ & 24 & 6 & 40320 & 720 \end{array}$$

$$\begin{array}{l} 4! \rightarrow 1 \times 2 \times 3 \times 4 \\ 3! \rightarrow 1 \times 2 \times 3 \\ 8! \rightarrow 1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 \times 8 \\ 6! \rightarrow 1 \times 2 \times 3 \times 4 \times 5 \times 6 \end{array}$$

sol $\rightarrow$ iterate and get min
ans = fact(min)

$$\rightarrow \begin{array}{l} TC \rightarrow O(N) \\ SC \rightarrow O(1) \end{array}$$

11 Subsequence

⇒ A subseq. can be generated from array
by removing 0 or more elements

arr[1] = { 3 2 ~~7~~ 6 ~~4~~ 8 ~~10~~ 9 }

↪ { 3 2 6 8 9 } → subseq.

↪ { ~~2~~ ~~9~~ ~~6~~ ~~8~~ }

↓
not a subseq.

↓
don't need to be
contiguous, order
should be
maintained

arr[4] = { 6 9 0 8 }

↓

{ 9 8 } ✓

{ 0 9 6 } ✗✗

Q3. Given an array return true if there exists any subsequence whose gcd = 1

ex $\Rightarrow$ 4 6 3 8 $\Rightarrow$ {4, 3, 8} $\rightarrow$ true

$$\downarrow$$

$$\text{gcd}(\text{gcd}(4, 3), 8)$$

$$\downarrow$$

$$\text{gcd}(1, 8)$$

$$\downarrow$$

$$1$$

ex $\Rightarrow$ 16 10 6 15 27 $\Rightarrow$ {10, 27} $\rightarrow$ ①

$$\text{gcd}(10, 27) \rightarrow 1 \quad \underline{\text{true}}$$

ex $\Rightarrow$ 6 12 3 18 $\Rightarrow$ false

ex $\Rightarrow$ 6 11 18 $\Rightarrow$ $\left. \begin{array}{l} (6, 11) \\ (11, 18) \\ (6, 11, 18) \end{array} \right\} \text{gcd} \rightarrow$

Obs

arr $\Rightarrow$ A $\checkmark$ B $\checkmark$ C $\checkmark$ D $\checkmark$ E $\checkmark$ F

gcd of full array $\Rightarrow$ gcd(A, B, C, D, E, F)

$\downarrow$

$$\text{gcd}(\underbrace{\text{gcd}(B, C, E)}_1, \text{gcd}(A, D))$$

$\downarrow$ $\checkmark$

assume $\rightarrow \text{gcd}(b, c, e) = 1$

$$= 1.$$

Som if gcd of entire array = 1
then there definitely exists a subseq
whose gcd = 1.

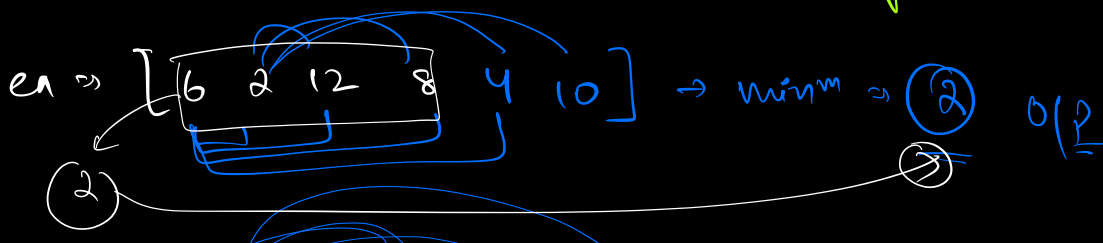
Todo $\Rightarrow$ find gcd of entire array $\Rightarrow$

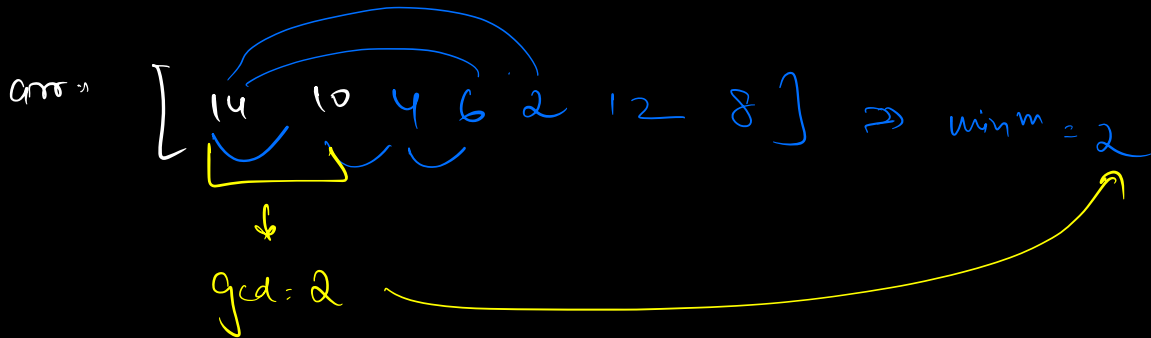
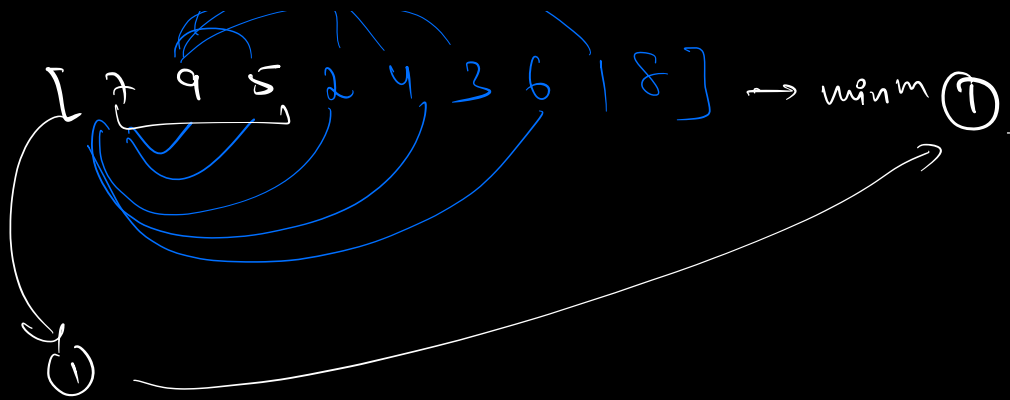
$$TC \Rightarrow O(N \log(\max(arr)))$$

$$SC \Rightarrow \underline{\underline{O(1)}}$$

$\Rightarrow$ Power Research and Facebook

Given N positive distinct elements in arr, choose
 $arr[i] \neq arr[j]$, such that $i \neq j$. If $abs(arr[i] - arr[j])$
is not present array, insert it ($abs(arr[i] - arr[j])$), keep
repeating the steps until no more insertions are possible.
Find the min^{im} value in the final array.





$\text{ans} = \text{min}$ of entire array.

$\Rightarrow \underline{\underline{\text{gcd of 1/p array}}}$

$\text{min} \rightarrow \text{gcd of array}$

$\text{max} \rightarrow \text{max element array}$

* find the length of final array.

Size of final array $\rightarrow$

